Team Blue – CS410

VibeQueue

Your Vibes, Managed – The Modern Jukebox for Local Spaces

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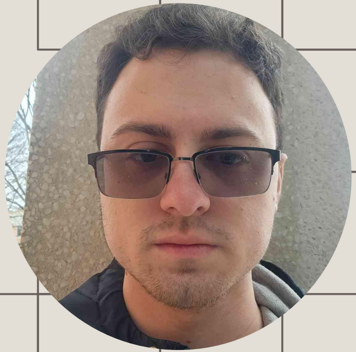
Team Bio

Lucas Hayes
Database Specialist

Lucas is a Senior at ODU majoring in Computer Science. He likes to read and listen to music.

Alanna Ivey
Documentation Specialist

Alanna is a Senior at ODU, majoring in Computer Science and Cybersecurity. She likes to try new foods and travel.



Spencer Hall
Software Developer

Spencer is a senior at ODU who is majoring in Computer Science. He likes to play games and watch sports.

Team Bio

Andrew Behre
Project Lead

Andrew Behre is a Junior at ODU who is majoring in Computer Science. He is a barista on campus and likes to skateboard.

Bea Eggleston
Webmaster

Bea Eggleston is a senior at ODU and is majoring in Computer Science. She enjoys programming and making art for games.

Dominick Rodriguez
Backend Specialist

Dominick Rodriguez is a Junior ODU majoring in Computer Science. He likes to read and listen to music.

Societal Problem

In the modern hospitality industry, atmosphere is important, but small cafes and restaurants struggle to manage their background music. A study by MCR data shows that 41% of customers will spend more time and money in a business if they like the music (2021). Current options are either expensive and hardware-heavy or lack customization. As a result, staff often handle requests manually, which is inefficient during busy times. At the same time, businesses miss out on customer engagement and potential revenue because there isn’t an affordable, controlled, and legal way for customers to interact with the music.

Who is affected

Our potential customer base and those who are impacted by this problem

1

Users

- Want influence over the atmosphere
- Prefer seamless, low-friction digital interaction

2

Customers

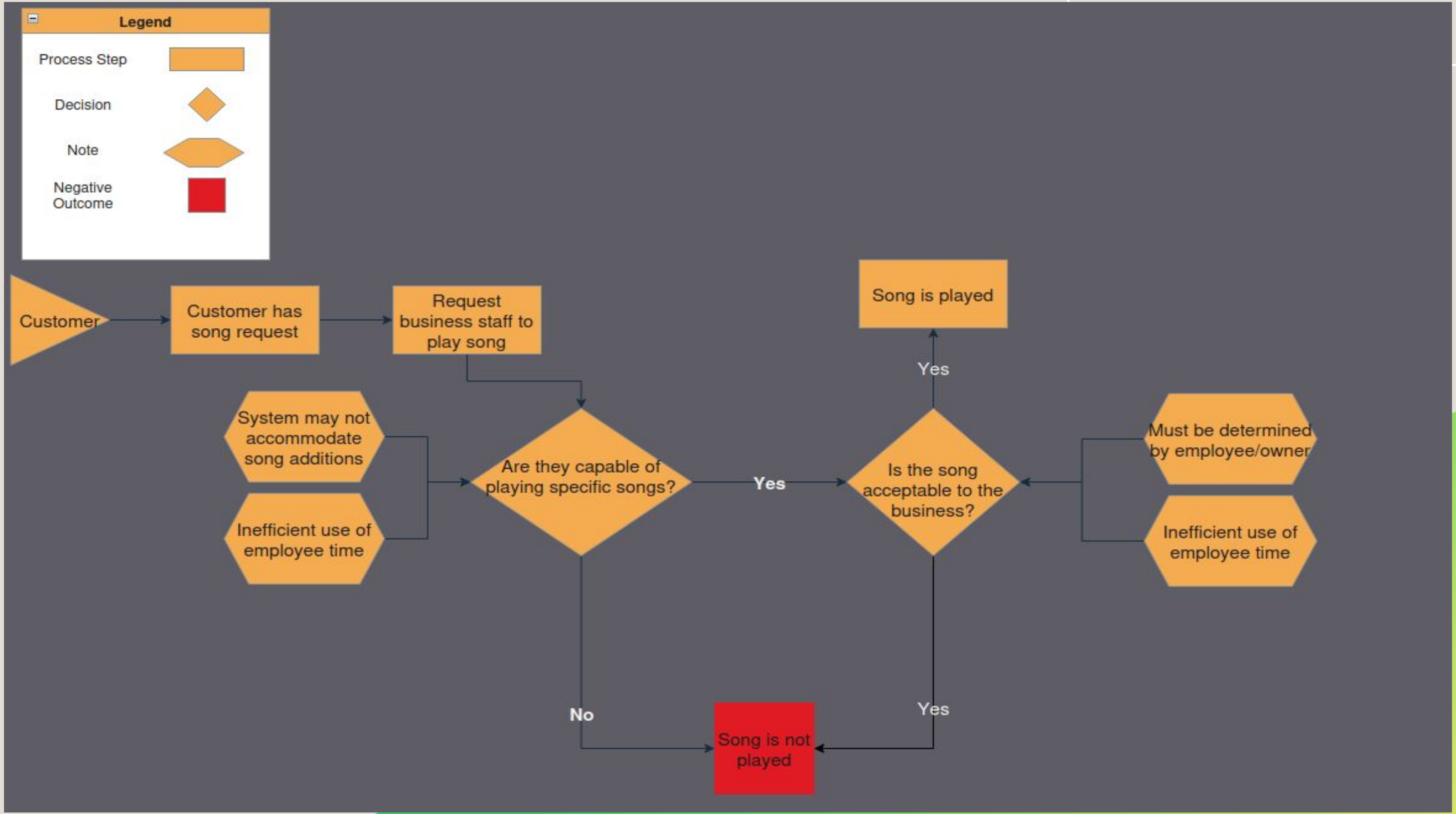
- Want increased engagement without staff disruption
- Need solutions that improve efficiency during peak hours

3

Stakeholders

Stakeholders are artists, record labels, and music platforms (iTunes, Spotify, etc.)

Current Process Flow



VibeQueue (Elevator Pitch)

Imagine walking into your favorite local coffee shop, scanning a QR code on your table, and instantly being able to influence the vibe of the room. VibeQueue is a software-based solution that replaces expensive hardware with a seamless digital ecosystem. By connecting the Spotify API with SoundMachine, we provide a fully licensed, legal, and interactive music experience for small businesses.

The Solution

- **A Web-Based Ecosystem:** VibeQueue automates music selection by providing a digital interface for customers to browse and request songs via QR codes.

- **Eliminates Manual Work & Hardware:** Removes the need for staff intervention and replaces expensive, proprietary hardware with a software-only solution.

- **Seamless API Integration:** Initial prototyping explored the Spotify API; however, strict commercial-use restrictions introduced legal risk. This risk is mitigated by using Soundcharts for music metadata and SoundMachine for fully licensed commercial playback.
- **Low-Cost & Interactive:** Offers an affordable "Vibe Management" tool tailored specifically for small-scale local spaces, differentiating from competitors with expensive physical consoles.

Problem Solution Mapping

Problem	Our Solution
Barrier to Enter: Commercial systems have a higher price than is feasible for many small businesses	Subscription based. VibeQueue is designed for small business groups operating on a more limited budget.
Limited User Experience: Rockbot has limited mobile access or is slower than customers want to use	Web-App: A Modern web-app allows for ease of customer use and high speed
Atmospheric Clash: Owners may be concerned that customers will play a song that doesn't fit	The Control Center: Owners have a dashboard that allows them to restrict or manipulate the qualities of the songs chosen

Automated Interaction (The "No-Staff" Gap)

Direct-to-consumer web interface allows for song requests without interrupting staff, resolving the "social friction" gap.

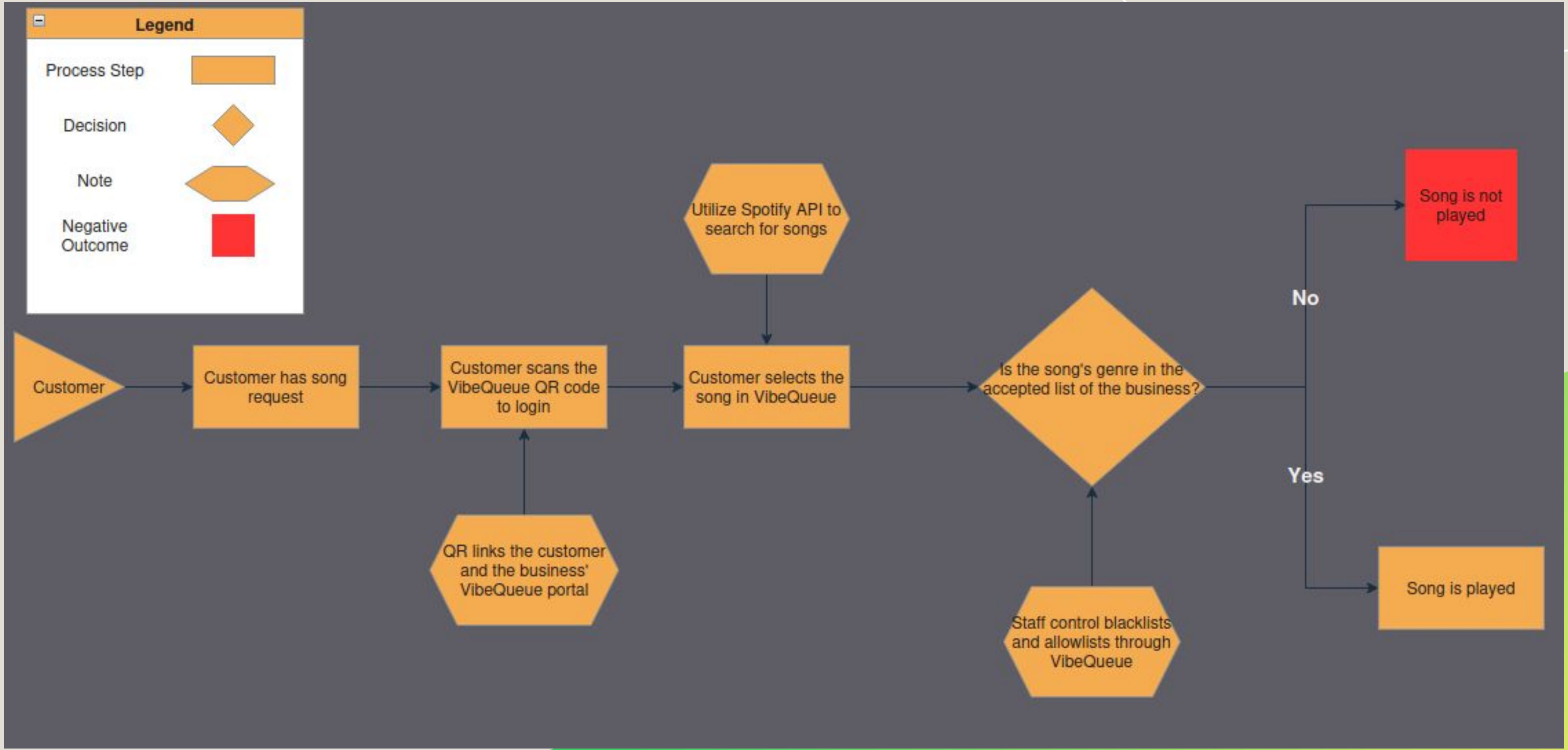
Hardware-Agnostic Deployment (The "Cost" Gap)

A software-only solution that runs on existing business devices (tablets/phones), removing the need for the expensive proprietary hardware required by TouchTunes or Rockbot.

Dynamic Vibe Moderation (The "Control" Gap)

Integrated filters and a "Business Control Center" allow owners to set genre limits and block explicit content automatically, ensuring the audio matches the establishment’s brand.

VibeQueue Solution Process Flow



Competition Matrix

	VibeQueue	TouchTunes	Rockbot
Project Stage	Prototype	Mature commercial product	Established commercial product
Target Market	Small local business	Large bars & restaurants	Medium/large business
Hardware Equipment	No specific hardware requirements	Proprietary jukebox required	Dedicated hardware recommended
Cost to Business	Low Cost	High upfront cost	Medium Cost
Best Use Case	Budget friendly/ Interactive spaces	High traffic venues	Curated Background Music

Major Functional Components

What software will be used (LAMP) to facilitate this project

1. **Soundcharts API**
Provides commercially safe music metadata and analytics

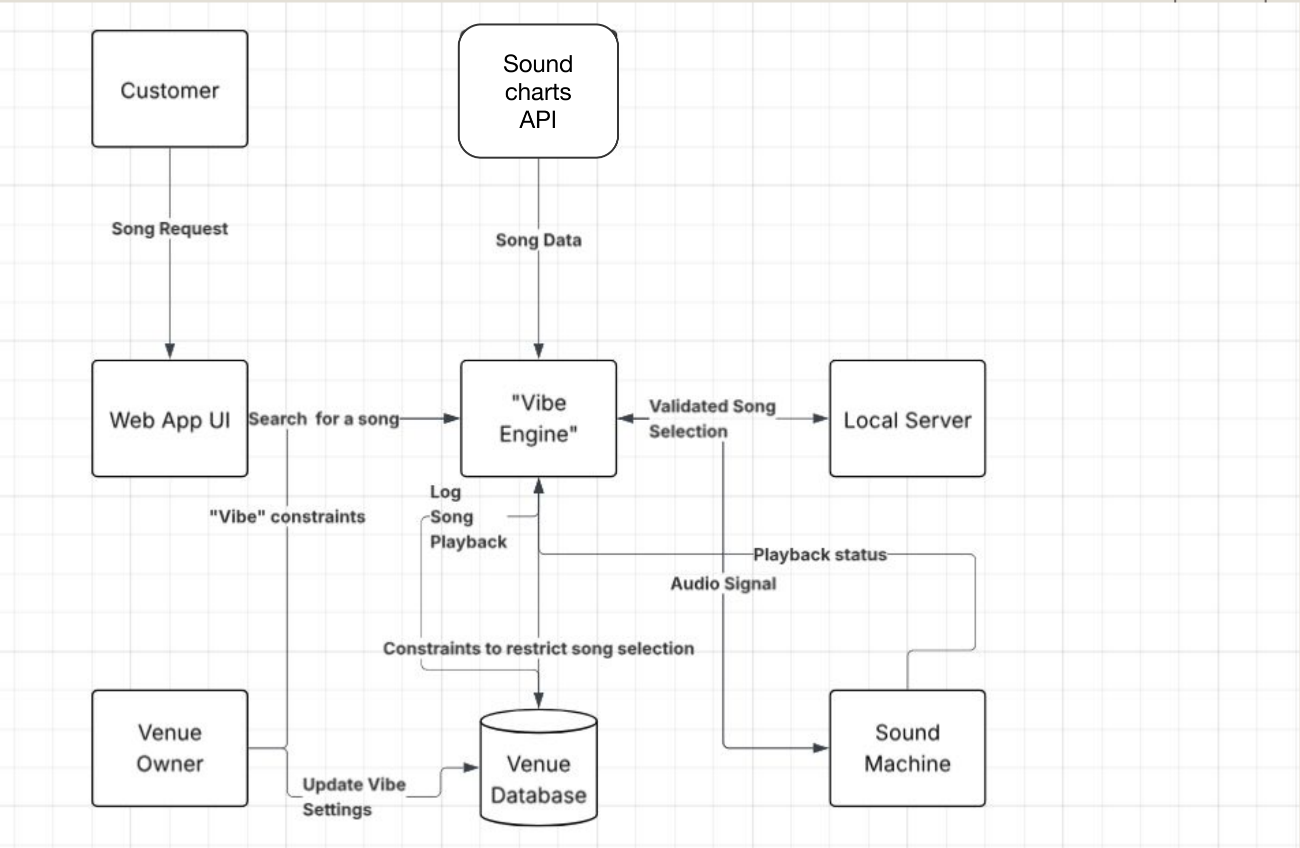
2. **Queue Manager**
Orders songs and applies priority logic

3. **Custom Song Selection Engine**
Built by the VibeQueue team to replace restricted streaming APIs
Enables "vibe" matching through genre, tempo, popularity, and mood data

4. **Recommendation Engine**
Suggest songs using Soundcharts metadata

MFCD Graph

Attached MFCD Diagram for visualization purposes



What It Will Do

1 Enable Frictionless Payment

- QR Code Access
- Song Browsing & Requests
- Influence the Atmosphere

2 Provide Owner & Staff Control

- The "Business Control Center"
  - A dedicated dashboard for owners
- Dynamic Vibe Moderation
- Queue Management
- User Management

3 Generate a Revenue Stream

- Pay-to-Request
- Automated Payment Processing
- Profit Sharing
  - The business keeps the majority of the revenue. Vibe Queue takes a small 5% cut to sustain development.

4 Ensure Legal Compliance

- Hardware Agnostic
  - Runs on devices the business already owns (staff phones, existing tablets)
- Licensed Streaming
  - Ensures all music played falls under legal commercial licenses (ASCAP, BMI, SESAC),

What It Will NOT Do

1

Play music without licensing

VibeQueue will not play any music that the business owner does not have the proper licensing to play.

Risks

#	Challenge	Response
1	TouchTunes is an established giant in this field	TouchTunes is designed for larger locations with higher budgets and resources. VibeQueue is intended for small businesses with minimal budget to spend on their music
2	Rockbot is an established company in this field	Rockbot lacks the convenience of a webapp and the customer facing aspects that are present in both TouchTunes and VibeQueue
3	Users selecting a song that doesn't fit the "vibe" of the shop	Staff at the shop will be able to utilize a "control center" which will allow them to limit or otherwise control the songs played at the location
4	Legal & API Compliance (Commercial Streaming Restrictions)	Pivoted from Spotify API to Soundcharts metadata + SoundMachine playback to ensure commercial licensing compliance

Risk Matrix

Likelihood	Med	High	Critical
	Low	Med	High
	Low	Low	Med
	Impact		

Risk Assessment

Risk Factor	Likelihood	Impact	Strategy
Competition (TouchTunes)	High	Medium	Target a different market (small businesses)
Competition(Rockbot)	Medium	Medium	Focus on user convenience
Brand Control	Low	High	Control center to protect the “vibe”
Legal & API Compliance	High	High	Pivot from Spotify API to Soundcharts metadata and SoundMachine playback to ensure full commercial licensing compliance

User Roles

Business Owner

- Can configure genre/mood filters
- Can manage queue rules
- Can view revenue
- Cannot bypass licensing rules

Staff

- Can reorder/remove songs
- Can block inappropriate requests
- Can assist customers
- Cannot change business settings

Customer

- Can scan QR code
- Can request songs
- Can pay for priority
- Cannot modify queue after submission

User Stories (Customer)

As a customer, I want to request a song that will influence the atmosphere.

Acceptance:

- Customer can search and select a song
- Song is validated against business filters
- Request is added to the queue
- UI confirms submission

As a customer, I want to pay to prioritize my request so that it plays sooner.

Acceptance:

- Customer can choose priority option
- Payment is processed successfully
- Request is placed higher in queue
- UI shows updated position

As a customer, I want to browse songs so that I can easily find music I like.

Acceptance:

- Songs are displayed by genre/vibe
- Search functionality is available
- Results load quickly
- Only allowed songs are shown

As a customer, I want to access the system via QR code so that I can quickly interact without downloading anything.

Acceptance:

- QR code opens web interface
- No login required (or simple session)
- Interface loads correctly on mobile
- User can immediately browse songs

User Stories (Owner/Staff)

As a business owner, I want to filter songs by genre and mood so that the music matches my brand.

Acceptance:

- Owner can set genre/mood restrictions
- Explicit content can be blocked
- Filters apply to all requests
- Disallowed songs are rejected automatically

As a business owner, I want to monitor revenue from song requests so that I can track business value.

Acceptance:

- Revenue dashboard is available
- Payments are recorded per request
- Data updates in real time
- Owner can view totals and trends

As a staff member, I want to override or remove songs so that I can manage inappropriate or disruptive requests.

Acceptance:

- Staff can remove songs from queue
- Queue updates immediately
- Action is reflected in UI
- Removed songs do not play

As a staff member, I want to view and manage the queue so that I can maintain a smooth music experience.

Acceptance:

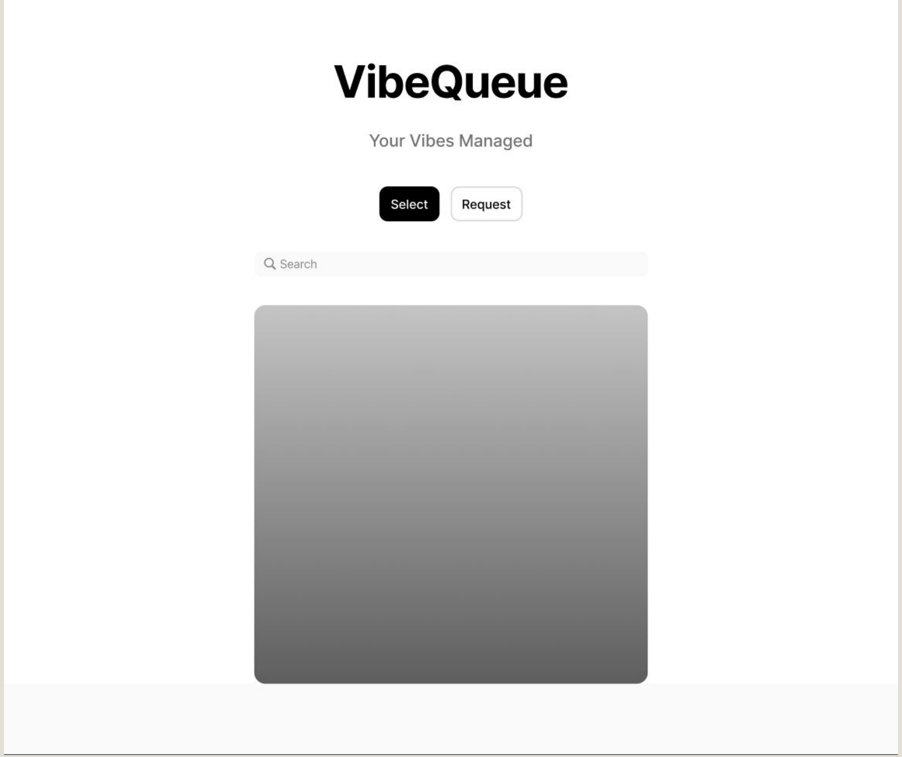
- Staff can view full queue
- Staff can reorder songs if needed
- Queue updates in real time
- Changes are reflected for all users

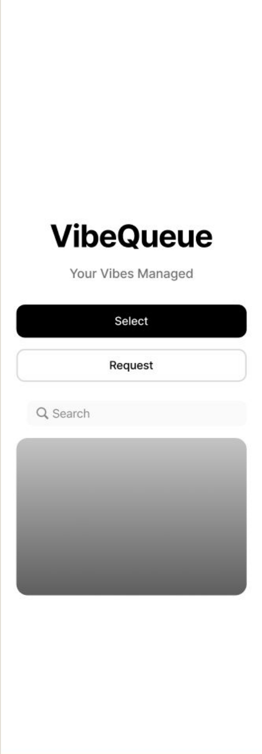
Feature Table

	Feature	Description	User roles
User Interaction	- QR code access - Song Browsing - Song Request - Pay-to-request	- Customers scan to access the system instantly - Browse songs using filters	Customer
Business Control	- Control center dashboard - Queue management - Content Filtering - Vibe moderation	- Central interface for managing music system - Approve, reorder, or remove songs - Block explicit or unwanted songs	Owner, Staff
System Features	- Recommendation engine - API integration - Licensed Playback	- Suggest songs based on vibe + data - Connects to Soundcharts + SoundMachine - Ensures legal music streaming	System

User Interface Mockups

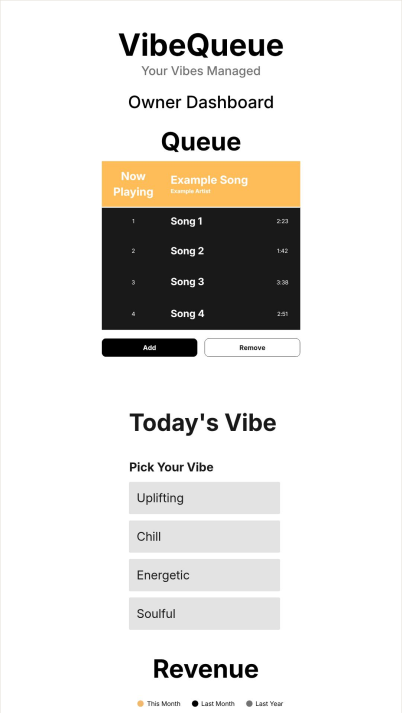
Customer Side

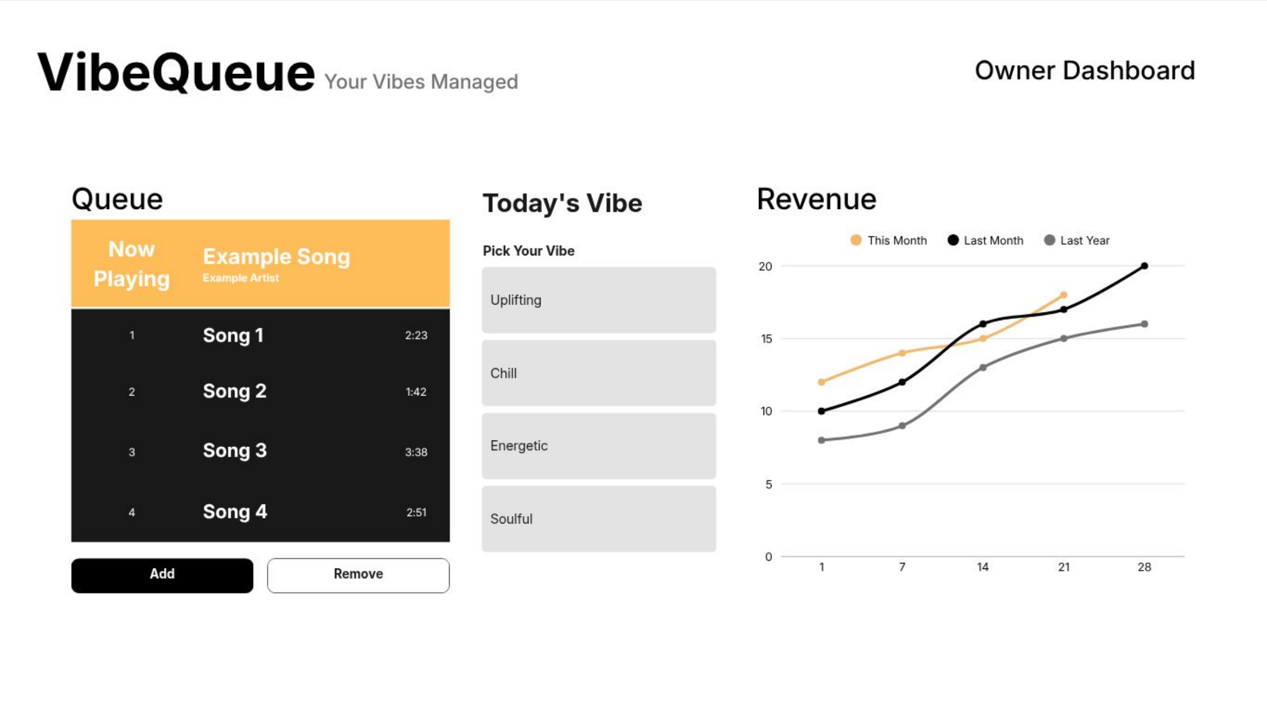




User Interface Mockups

Business Side





Work Breakdown Structure

This is a visual breakdown of everything that will need to be built, set up, designed, or tested during development.

Initial Sprint Breakdown

Seven 2-week sprints — from foundation to pilot launch

Sprint 1
Foundation

Repo, CI/CD, hosting (AWS/Render); SQL schema for Users, Venues, Songs, Queue, Transactions.

Sprint 2
Auth & Roles

Owner / Staff / Customer roles, secure login, password hashing, session handling.

Sprint 3
Vibe Match Engine

Soundcharts API integration, metadata cache, genre/tempo/mood tagging, vibe-match algorithm.

Sprint 4
Queue & Playback

Queue Manager with priority logic, SoundMachine playback handoff, real-time queue updates.

Sprint 5
Customer Web App

QR-code entry, mobile-first browse/search/request UI, filter enforcement, request confirmation.

Sprint 6
Control Center

Owner dashboard: vibe filters, explicit-content blocks, queue overrides, revenue analytics.

Sprint 7
Payments & Pilot

Stripe pay-to-prioritize, profit-share reporting, end-to-end QA, pilot deployment at partner cafe.

Algorithms

The Recommender Analogy

Social platforms learn what an individual likes from clicks, watch time, and skips. VibeQueue applies the same idea — but the listener is the venue, not a person.

How the analogy maps

- **User profile** → Venue vibe vector
- **Engagement signals** → Queue / skip / override events
- **Recency & repetition** → Repeat-play penalty per shift
- **"For You" feed** → Auto-curated request menu

Song Selection & Matching

Catalog Reconciliation. Pulls metadata from Soundcharts (artist, album, ISRC) and locates the equivalent track in SoundMachine's licensed catalog.
Fuzzy Fallback. On miss, scores candidates with token-set similarity + artist match; picks the closest legal match above a threshold.

Vibe Match Engine

Feature Vector. Each song encoded across genre, BPM, energy, valence, danceability, popularity, era, and explicit flag.
Venue Profile. Owner sets a target vector per shift. Songs ranked by weighted cosine similarity; hard filters gate the result.

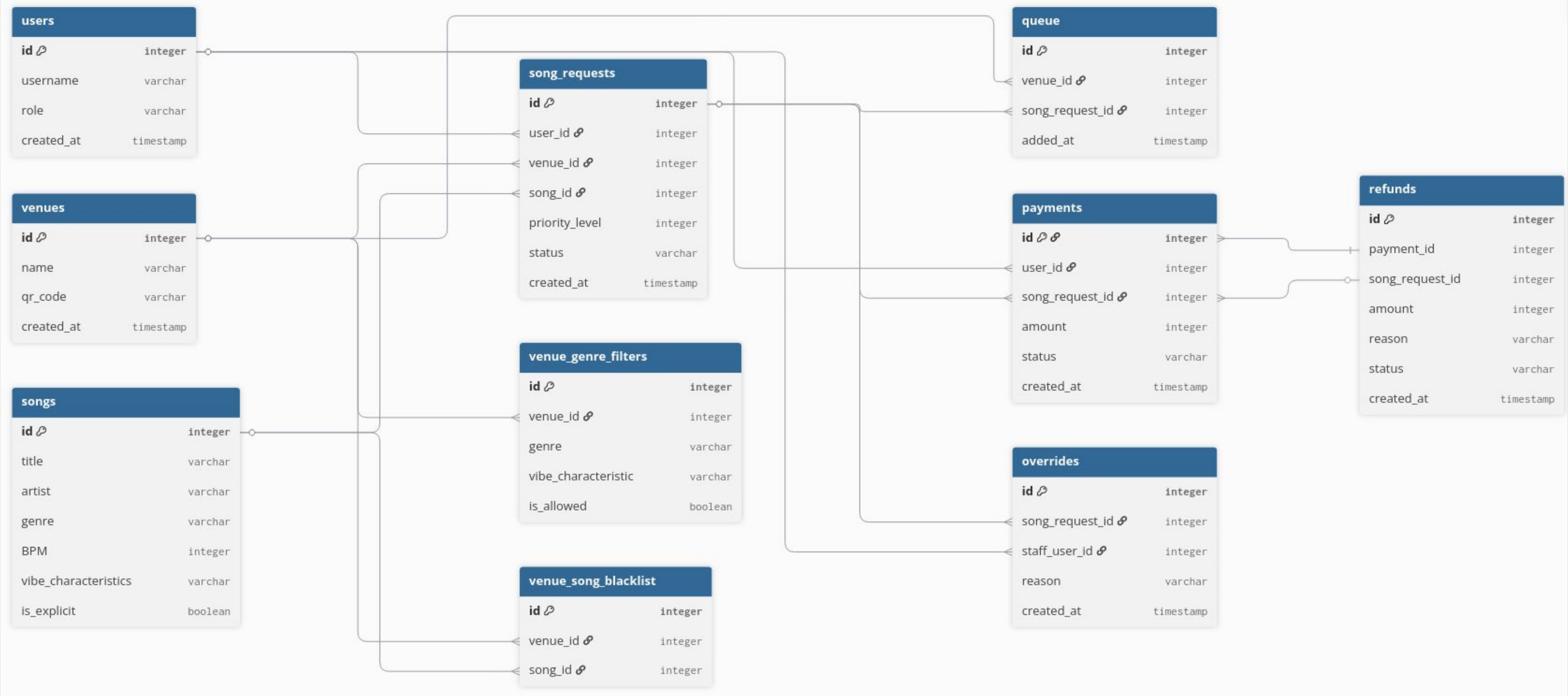
Adaptive Vibe Learning

Implicit Feedback. Queue accepts, skips, requests-per-shift, and owner overrides nudge the venue's vibe vector over time.
Recommender-Style Loop. Same mechanism a "For You" feed uses to learn a person — applied here to learn the venue's atmosphere.

Dependencies

Soundcharts API	This is needed to access and the songs as well as collect and manage data associated
QR code payment system	Needs to have a system in place to allow payment through a QR code (Stripe)
SoundMachine	Is used to actually play the songs and control the playing of the music within the locations. It also handles all of the licensing for any music to be played.

Database Schema



Data Management Approach

1 Relational Storage Architecture (SQL)

- **Structured Data:** Utilizing an SQL database to effectively manage relationships between Users, Venues Song Queues, and Transactions
- **Hosting & Scalability:** Deployed via cloud hosting (AWS/Render) to ensure high availability and seamless scaling during peak venue hours.

2 API Integration & Data Flow

- **Metadata Caching:** Song metadata (genre, tempo, explicit flags) retrieved from the Soundcharts API is temporarily cached to minimize latency and reduce redundant API calls.
- **Stateless Playback:** VibeQueue manages the queue logic, while actual audio streaming is offloaded entirely to SoundMachine, avoiding heavy local audio file storage and copyright liabilities.

3 Privacy & Security

- **Payment Tokenization:** All financial transactions are processed securely through Stripe. VibeQueue does not store raw credit card data on its servers.
- **Data Protection:** User passwords and sensitive owner credentials are encrypted using industry-standard hashing algorithms before storage

4 Data Integrity & Reliability

- **Transactional Safety:** Employing database transactions to ensure atomicity meaning a customer is only charged if their song request successfully passes the Vibe Engine and enters the queue.
- **Disaster Recovery:** Routine automated database backups to prevent data loss regarding venue settings, user accounts, and transaction history.

Conclusion and Key Points

- Music is a critical factor of customer satisfaction and loyalty
  - Customers are 39% more likely to return to a business if they enjoy the music (MCR Data, 2021)
  - 41% of Consumers will spend more time/money in a business if they like the music (MCR Data, 2021)
- The existing competition does not cater to smaller businesses or spaces.
  - Higher costs/upkeep
  - Specific hardware requirements
- VibeQueue is a low-cost easy system that can meet the needs of the customer.
  - Cheap
  - Easy to use
- VibeQueue optimizes the current logistical overhead created by other competitors (RockBot and TouchTunes) as an open platform to expand to smaller businesses
  - Hardware-agnostic
  - Budget-friendly
  - Web-accessible

Glossary and References

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Questions?

Help us fine-tune the vibe!